

Analysis 5: Subgroup Fairness Analysis

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0.1 Overview

Subgroup discrimination and calibration of the Analysis-1 nomogram's out-of-sample risk are assessed here, stratified by sex and age band. For each subgroup the sample size, event rate, AUROC, and calibration-in-the-large (mean predicted risk versus observed rate) are reported. External validation on eICU is deferred (no local database available).

0.2 Load Libraries

```
suppressPackageStartupMessages({ library(pROC) })
```

0.3 Locate and Read Data

```
find_data <- function(f) {  
  cands <- c(file.path("../processed_data", f),  
             file.path("processed_data", f), f)  
  hit <- cands[file.exists(cands)][1]; if (is.na(hit)) stop("cannot find ", f); hit  
}  
OUT <- dirname(find_data("10_analysis1_oos_pred.parquet"))  
  
pred <- as.data.frame(arrow::read_parquet(find_data("10_analysis1_oos_pred.parquet")))  
lm6 <- as.data.frame(arrow::read_parquet(find_data("08_landmark_h6.parquet")))[, c("stay_id", "a  
d <- merge(pred, lm6, by="stay_id")  
d$age_band <- cut(d$age, c(-Inf, 50, 65, 80, Inf),  
                 labels=c("<50", "50-64", "65-79", "80+"))
```

0.4 Define Subgroup Metrics Function

For each subgroup, AUROC, mean predicted risk, observed event rate, and their ratio (calibration-in-the-large) are computed. A calibration ratio of 1.0 indicates that the average prediction matches the

true event rate; values above 1 indicate over-prediction, values below 1 under-prediction.

```
grp_metrics <- function(sub, label) {  
  y <- sub$onset_within_h; p <- sub$risk_en  
  au <- tryCatch(as.numeric(auc(roc(y, p, quiet=TRUE, direction="<"))), error=function(e) NA)  
  data.frame(subgroup=label, n=nrow(sub), event_rate=round(mean(y),4),  
             auroc=round(au,4), mean_pred=round(mean(p),4),  
             obs_rate=round(mean(y),4),  
             calib_ratio=round(mean(p)/mean(y),3))  
}
```

0.5 Compute Subgroup Metrics and Write Results

Fairness metrics are computed for the full cohort and for each sex and age-band subgroup, then saved to analysis5_subgroups.csv.

```
rows <- list(grp_metrics(d, "ALL"))  
for (g in sort(unique(d$gender)))  
  rows[[length(rows)+1]] <- grp_metrics(d[d$gender==g,], paste0("sex=",g))  
for (b in levels(d$age_band))  
  rows[[length(rows)+1]] <- grp_metrics(d[d$age_band==b,], paste0("age=",b))  
tab <- do.call(rbind, rows)  
print(tab, row.names=FALSE)
```

##	subgroup	n	event_rate	auroc	mean_pred	obs_rate	calib_ratio
##	ALL	48829	0.0682	0.7752	0.0682	0.0682	1.000
##	sex=F	21519	0.0629	0.7615	0.0631	0.0629	1.003
##	sex=M	27310	0.0724	0.7838	0.0722	0.0724	0.998
##	age=<50	9680	0.0584	0.7561	0.0616	0.0584	1.056
##	age=50-64	14083	0.0670	0.7845	0.0634	0.0670	0.945
##	age=65-79	15960	0.0690	0.7771	0.0680	0.0690	0.985
##	age=80+	9106	0.0790	0.7691	0.0829	0.0790	1.050

```
write.csv(tab, file.path(OUT, "16_analysis5_subgroups.csv"), row.names=FALSE)  
cat("\nAnalysis 5 (fairness) complete. External eICU validation deferred (no data).\n")
```

##

Analysis 5 (fairness) complete. External eICU validation deferred (no data).